

Advancing GeoAI for Disaster Resilience:

From Damage Assessment to Responsible Decision Support

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Committee members:

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10.22.2026@ Proposal Defense, O&M 807

"Seeing, describing, and governing disasters — beyond the boundaries of the screen, let the real world become the stage for intelligence." --Yifan Yang

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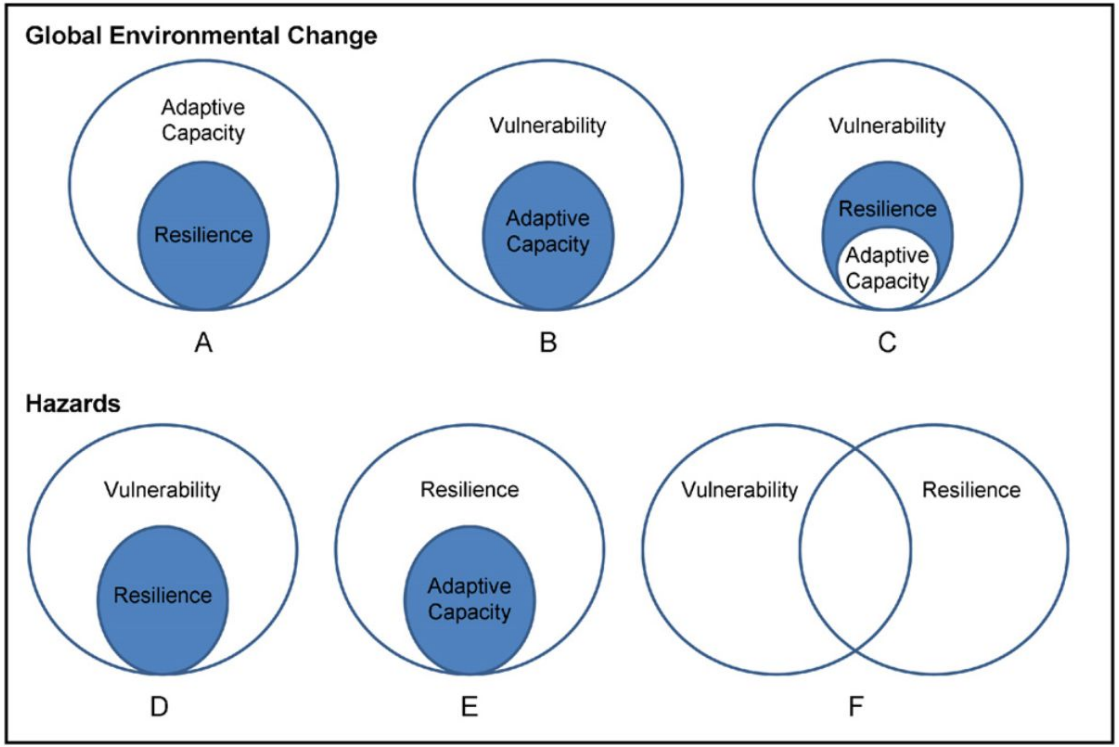
Outline

- Background
- Current Challenges
- Conceptual Framework
- Objectives
- Case Studies
- Publications and Awards
- Research Schedule

Background

Disaster resilience is a geographical question

(1) Rooted in space: resilience is place-based, shaped by local hazard exposure, social vulnerability, and human–environment interactions (Cutter et al., 2008).

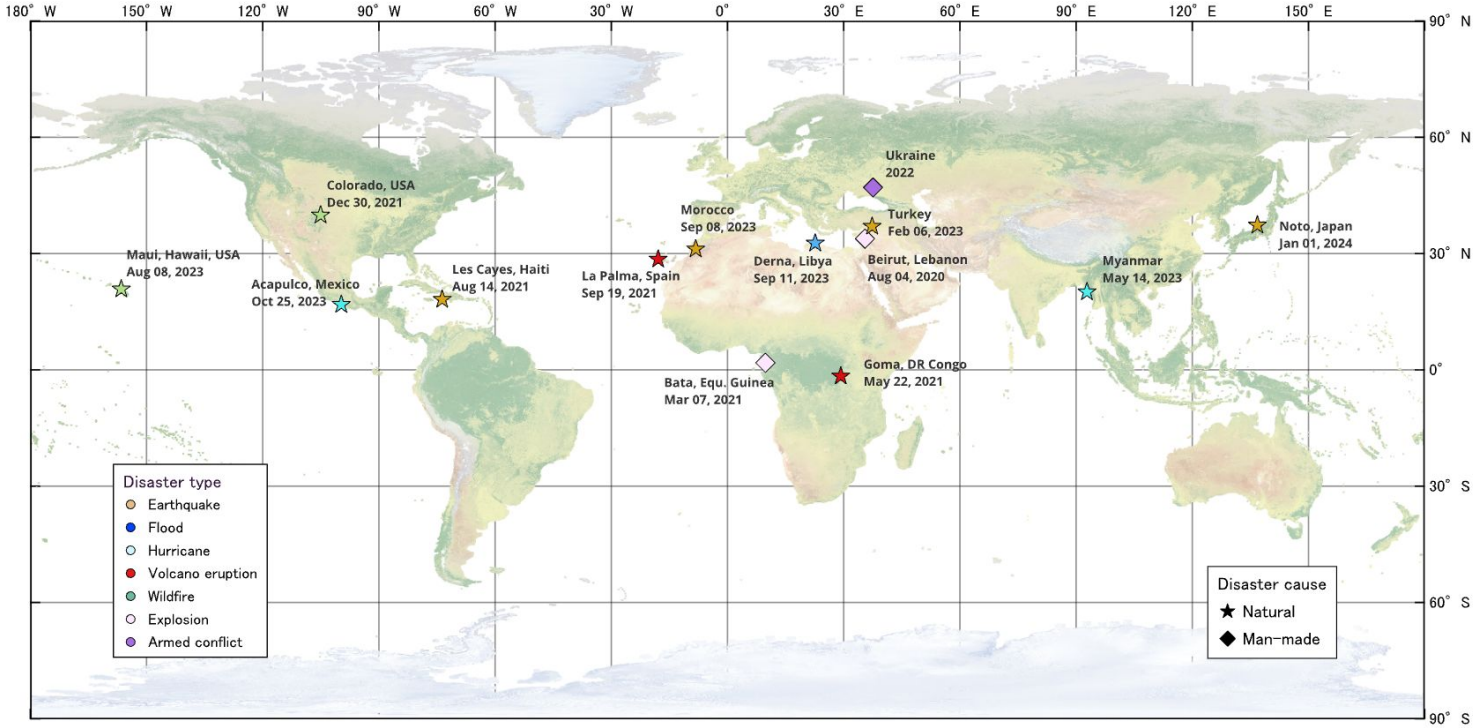


Cutter, S. L., Barnes, L., Berry, M., Burton, C., Evans, E., Tate, E., & Webb, J. (2008). A place-based model for understanding community resilience to natural disasters. Global Environmental Change, 18(4), 598–606.

Background

Disaster resilience is a geographical question

(2) Diverse across the globe: disasters strike different places with different hazards and impacts. The 14 BRIGHT events alone span five continents and seven disaster types, both natural and man-made (Chen et al., 2025).

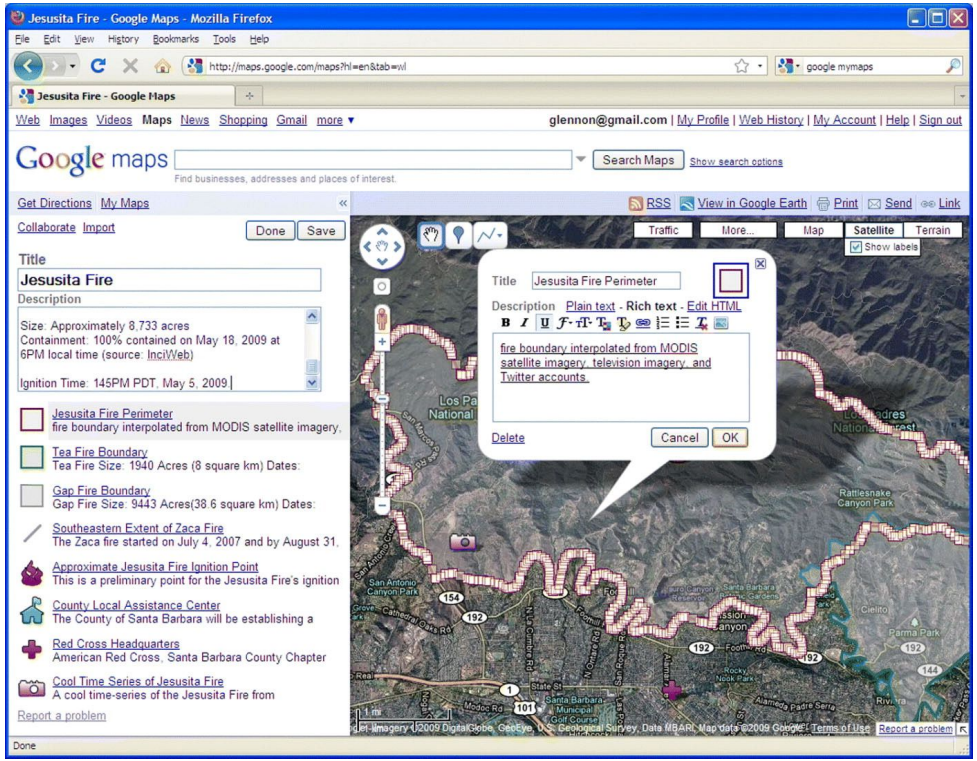


Chen, H., Song, J., Dietrich, O., Broni-Bediako, C., Xuan, W., Wang, J., ... & Yokoya, N. (2025). BRIGHT: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response. Earth System Science Data, 17(11), 6217-6253.

Background

Disaster resilience is a geographical question

(3) A geographic inquiry: beyond what happened, we ask where, how severe, who was affected, and how to allocate resources (Goodchild & Glennon, 2010).



Goodchild, M. F., & Glennon, J. A. (2010). Crowdsourcing geographic information for disaster response: a research frontier. International Journal of Digital Earth, 3(3), 231-241.

Background

Disaster resilience is a geographical question

(4) A cycle in time: resilience is the ability to prepare for, absorb, recover from, and adapt to adverse events (National Academies et al., 2012). Each phase, from preparedness to response, recovery, and mitigation, depends on timely, place-specific information.



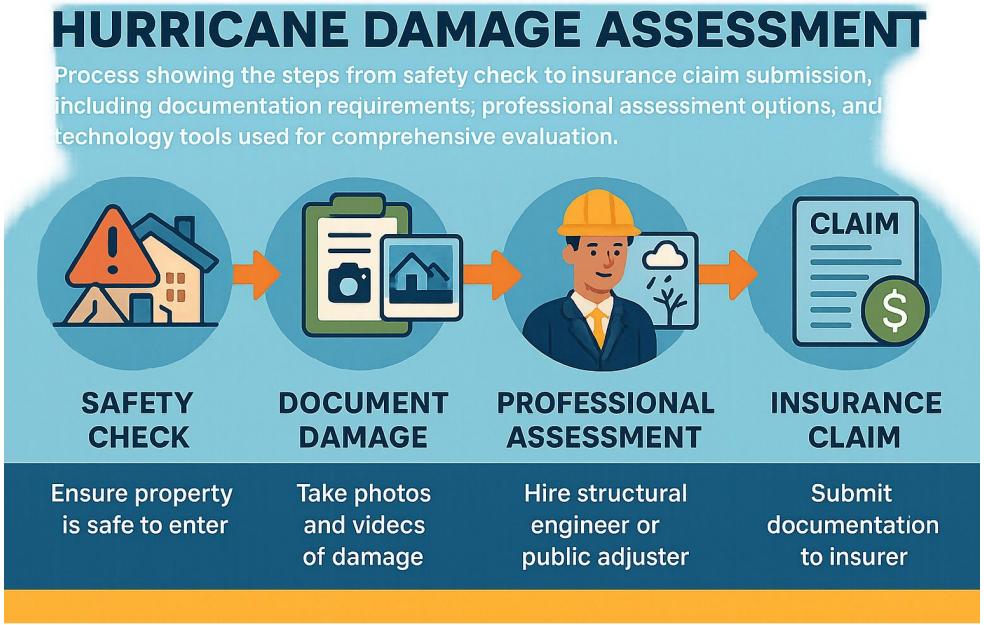
Image generated by ChatGPT

National Academies, Policy, Global Affairs, Committee on Science, Public Policy, & Committee on Increasing National Resilience to Hazards. (2012). Disaster resilience: A national imperative. National Academies Press.

Background

Disaster assessment is a critical task for building resilience

Core tasks: acquire imagery (remote sensing + street view) → classify severity → generate reports → map impacts rapidly (FEMA, 2016).





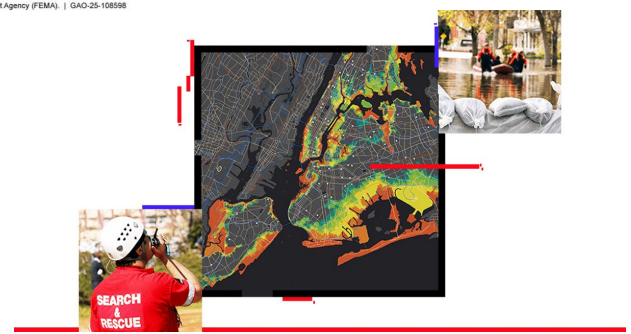
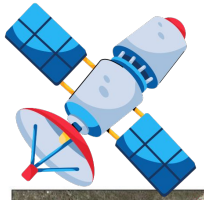


Image source: Insurance Claim Recovery Support, hurricane damage assessment workflow. · FEMA (2016). Damage Assessment Operations Manual: A Guide to Assessing Damage and Impact. U.S. Department of Homeland Security.

Background

Image data is central to disaster assessment: two complementary views



Overhead view — remote sensing

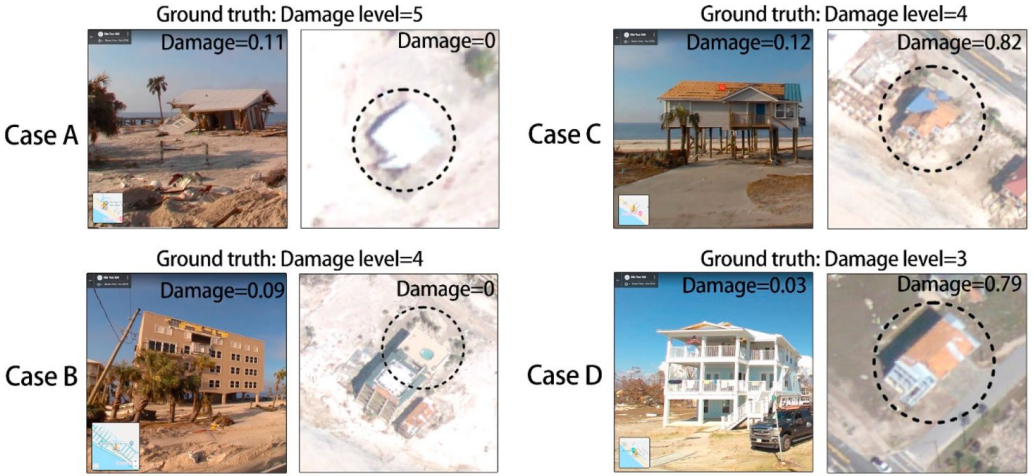
Satellite and aerial imagery: wide coverage and roof-level damage (e.g., xBD benchmark; Gupta et al., 2019).





Ground-level view — street-view imagery

Facade-level detail of structures, debris, and flooding, captured by survey vehicles, panoramic cameras, and volunteers (Zhai & Peng, 2020).



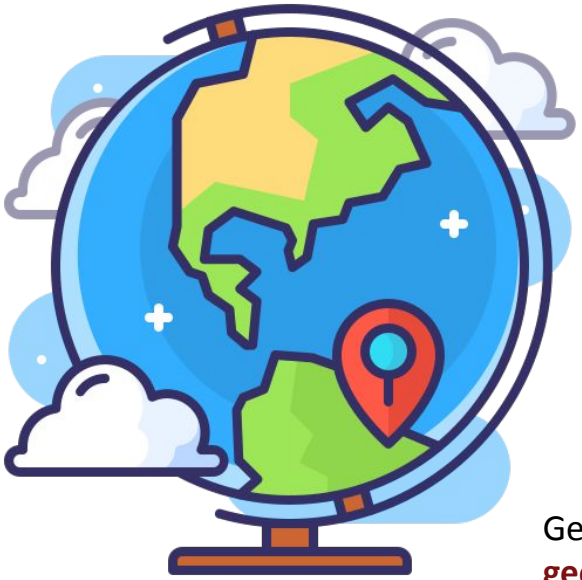
Left: Hauptman, L., Mitsova, D., & Briggs, T. R. (2024). Hurricane Ian damage assessment using aerial imagery and LiDAR: A case study of Estero Island, Florida. Journal of Marine Science and Engineering, 12(4), 668.

Right: Zhai, W., & Peng, Z. (2020). Damage assessment using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. Applied Geography, 123, 102252.

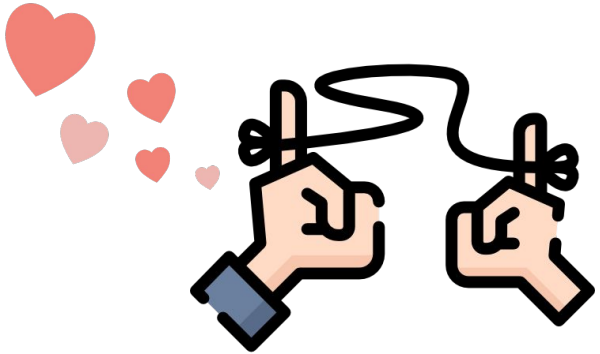
Gupta, R., Hosfelt, R., Sajeev, S., Patel, N., Goodman, B., Doshi, J., Heim, E., Choset, H., & Gaston, M. (2019). xBD: A dataset for assessing building damage from satellite imagery. CVPR Workshops.

Background

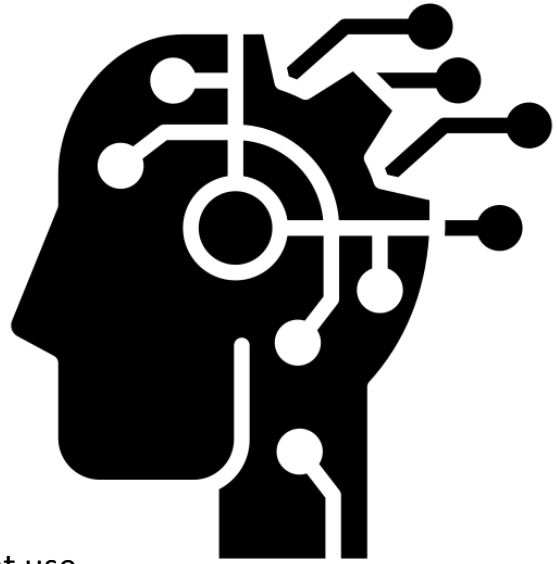
The engine: GeoAI for disaster resilience



Geography



Artificial Intelligence



GeoAI refers to **spatially explicit AI** methods that use **geographic data and spatial context** to **create** intelligent geographic information, knowledge, and services.

Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. International Journal of Geographical Information Science, 34(4), 625-636.

Background

Three families of AI models, three roles in GeoAI



Discriminative models — task-specific classifiers

"Is this building damaged? Mild, moderate, or severe?"

+ Accurate on the task they are trained for

− Need labels for every event; stay confident even when wrong



Generative models — foundation and vision-language models

"Why is this location severely damaged? What should responders do here?"

+ Zero- and few-shot generalization across hazards and regions

− Usually less accurate than tuned classifiers; may hallucinate



Agentic AI = Large Language Models (LLMs) + tools + context

"Plan the analysis, call the spatial tools, and show the evidence behind every claim."

+ Combines both families: classifiers for accuracy, foundation models for reasoning

+ Calls GIS tools and live data, so answers rest on computed evidence, not memory

+ Extends beyond damage assessment to the whole resilience cycle

Bommasani, R., et al. (2021). On the opportunities and risks of foundation models. arXiv:2108.07258. · Wang, L., et al. (2024). A survey on large language model based autonomous agents. Frontiers of Computer Science, 18(6), 186345. · Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. ICML.

Background — one-sentence summary

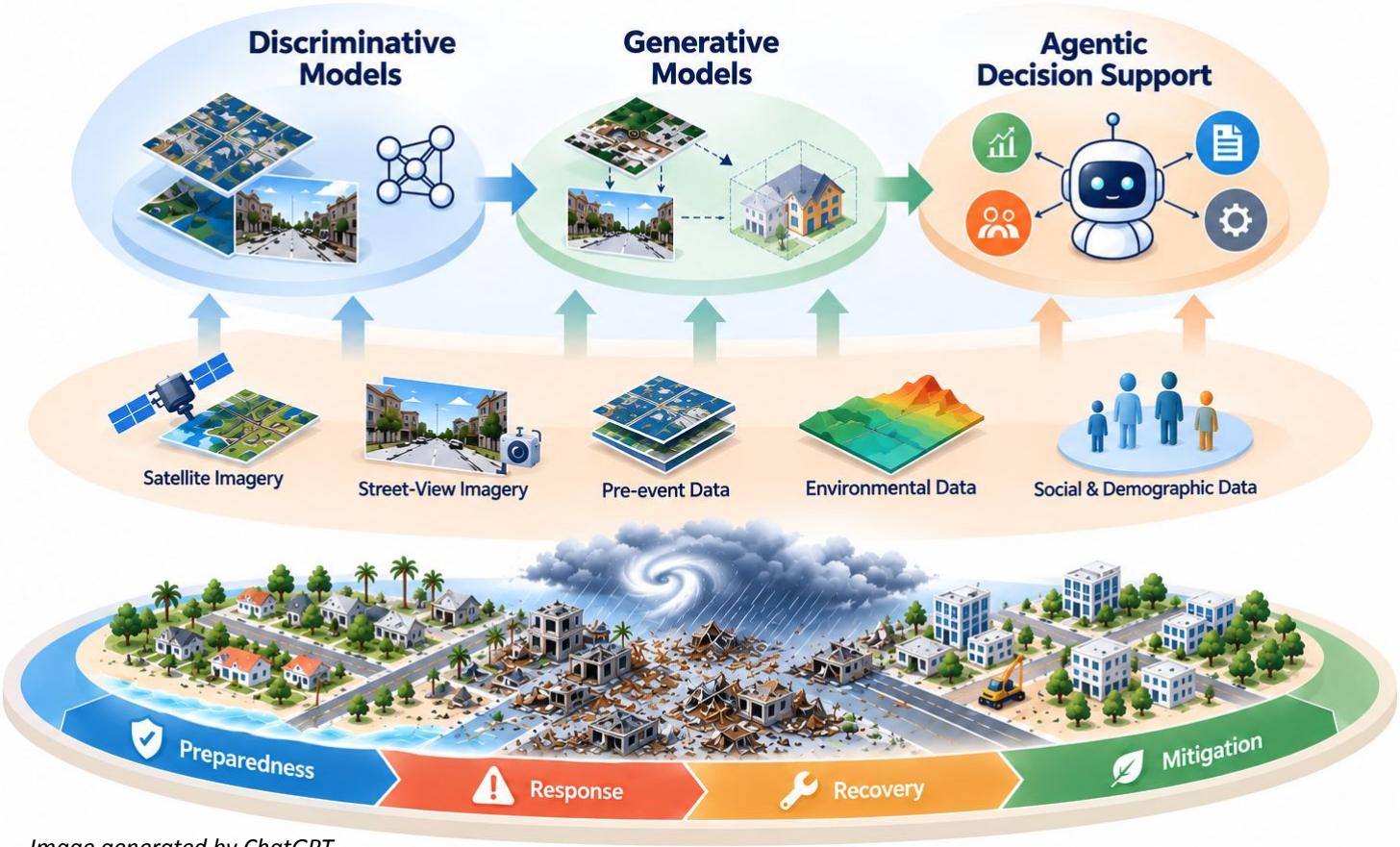


Image generated by ChatGPT

Advancing GeoAI along three evolving model families — discriminative, generative, and agentic — to move from damage assessment toward responsible decision support across the whole disaster resilience cycle.

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Current Challenges

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1 **Discriminative assessment is not yet accurate or reliable enough.**

Models that see only one post-event view misclassify damage and stay confident when wrong.

How to see better?

2 **Specialized models do not generalize across events.**

Each new disaster needs new labels and retraining, which is too slow for response.

How to respond faster?

3 **Assessment alone does not close the resilience loop.**

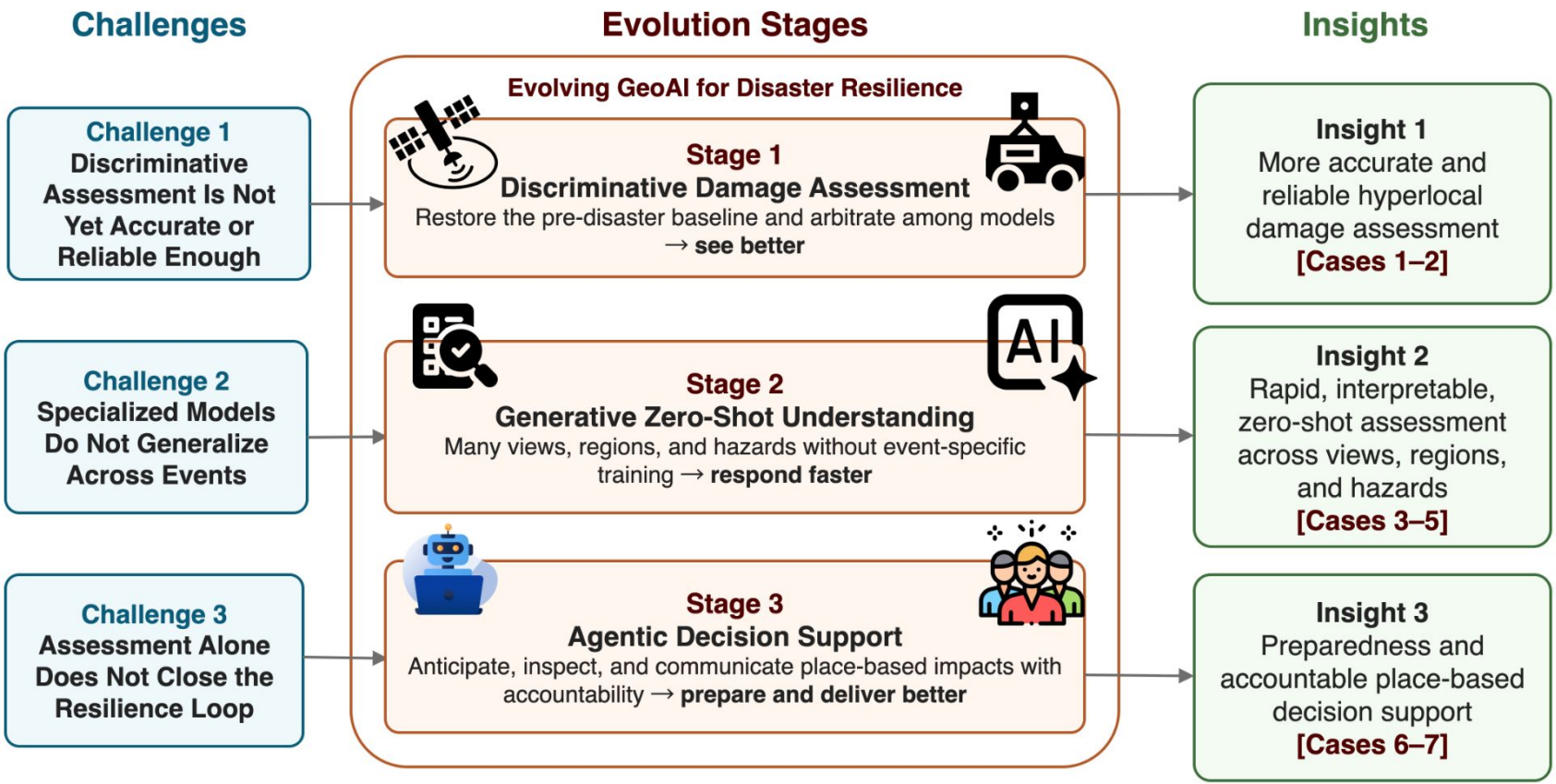
A damage label or map is not a decision; each phase needs actionable, accountable guidance.

How to prepare and deliver better?



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Conceptual Framework



Objectives

Objective 1 → *addresses Challenge 1 · Cases 1–2*

Improve accuracy and reliability by fusing different views and arbitrating among multiple models.

See better

Objective 2 → *addresses Challenge 2 · Cases 3–5*

Enable zero-shot assessment that transfers across hazards and regions without new labels or retraining.

Respond faster

Objective 3 → *addresses Challenge 3 · Cases 6–7*

Build accountable agentic GeoAI that turns assessment into evidence-backed action before, during, and after disasters.

Prepare and deliver better



STAGE 1 · OBJECTIVE 1

See better

Challenge 1 · Discriminative assessment is not yet accurate or reliable enough

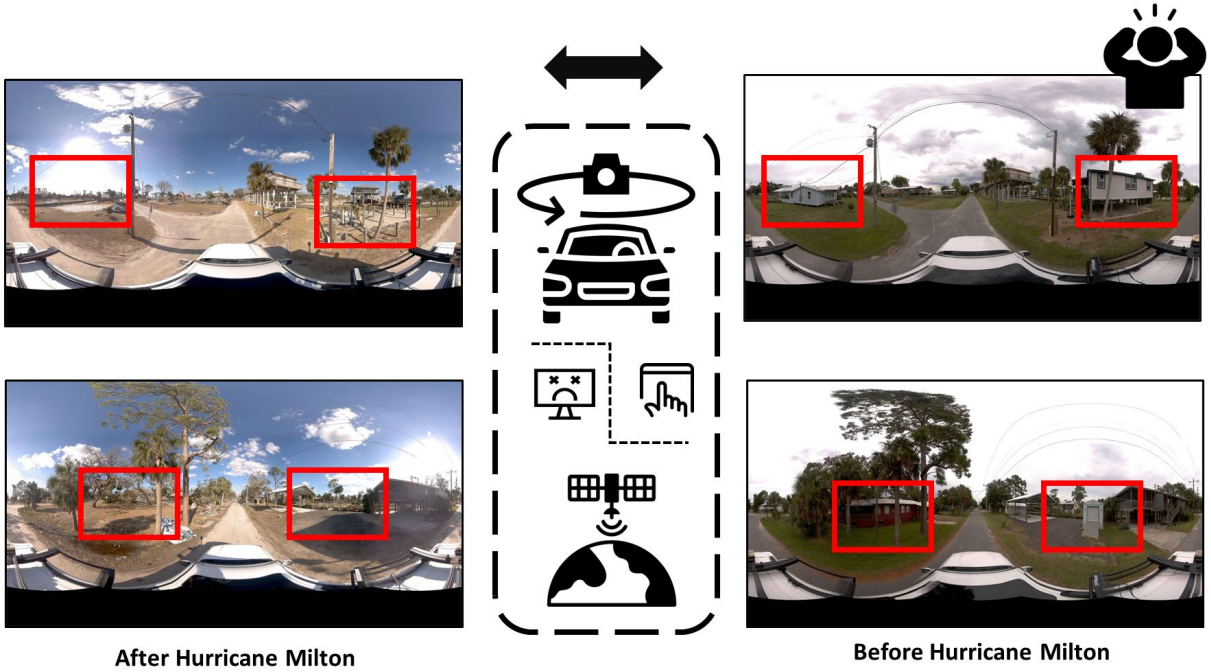
Improve accuracy and reliability by fusing pre- and post-event views and arbitrating among multiple models.

Case 1	Hyperlocal damage assessment from bi-temporal street-view imagery	Published · CEUS 2025
Case 2	DamageArbiter: multimodal arbitration for damage assessment	Under review · Natural Hazards

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Case 1:

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)



After Hurricane Milton

Before Hurricane Milton

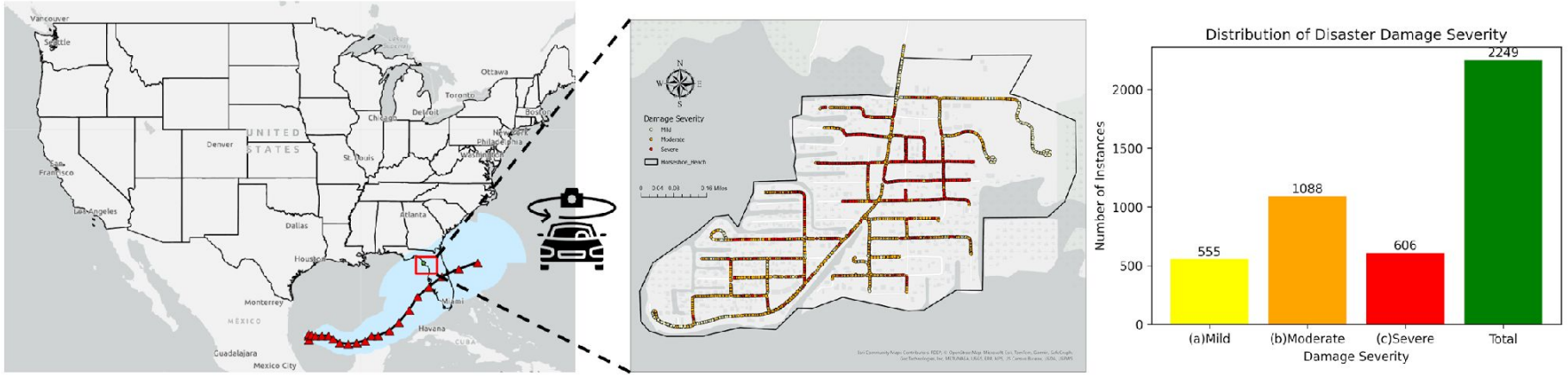
Research questions:

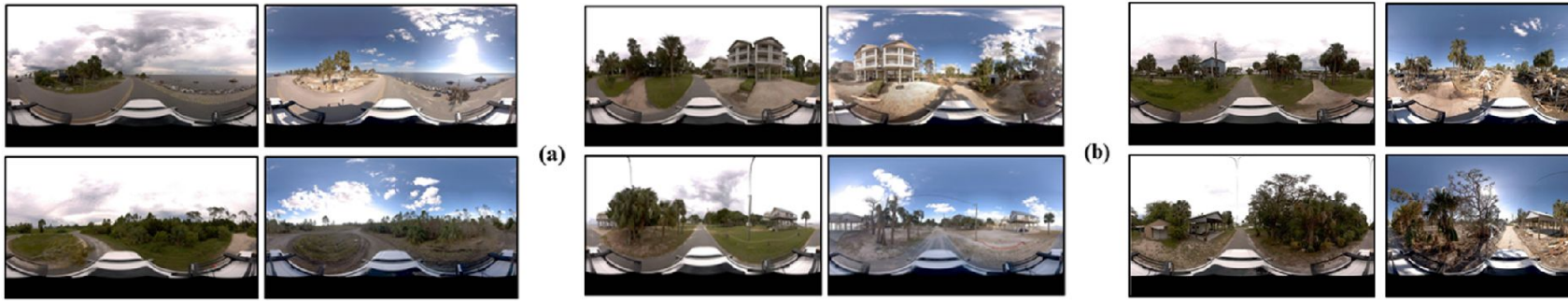
(1) Can pre-disaster street-view imagery help us classify disaster damage more accurately?

(2) How can we integrate pre- and post-disaster street-view imagery to enable rapid, hyperlocal damage assessment?

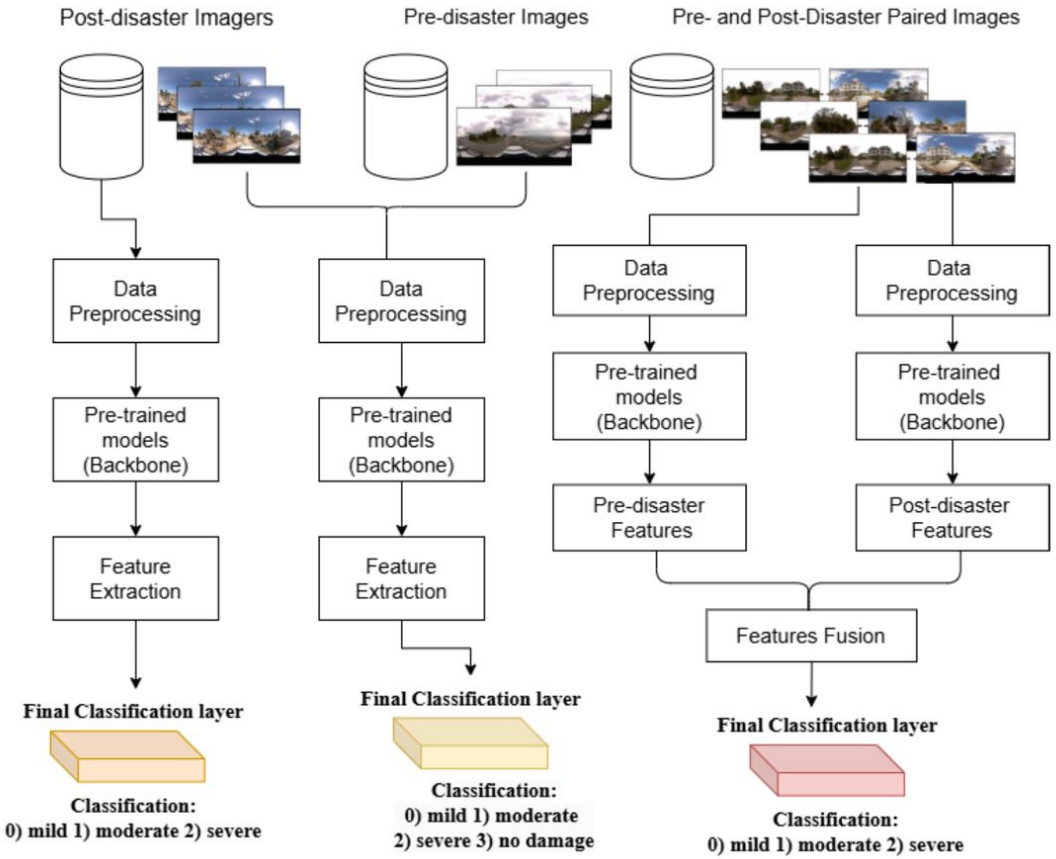
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Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (study area and dataset)





Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



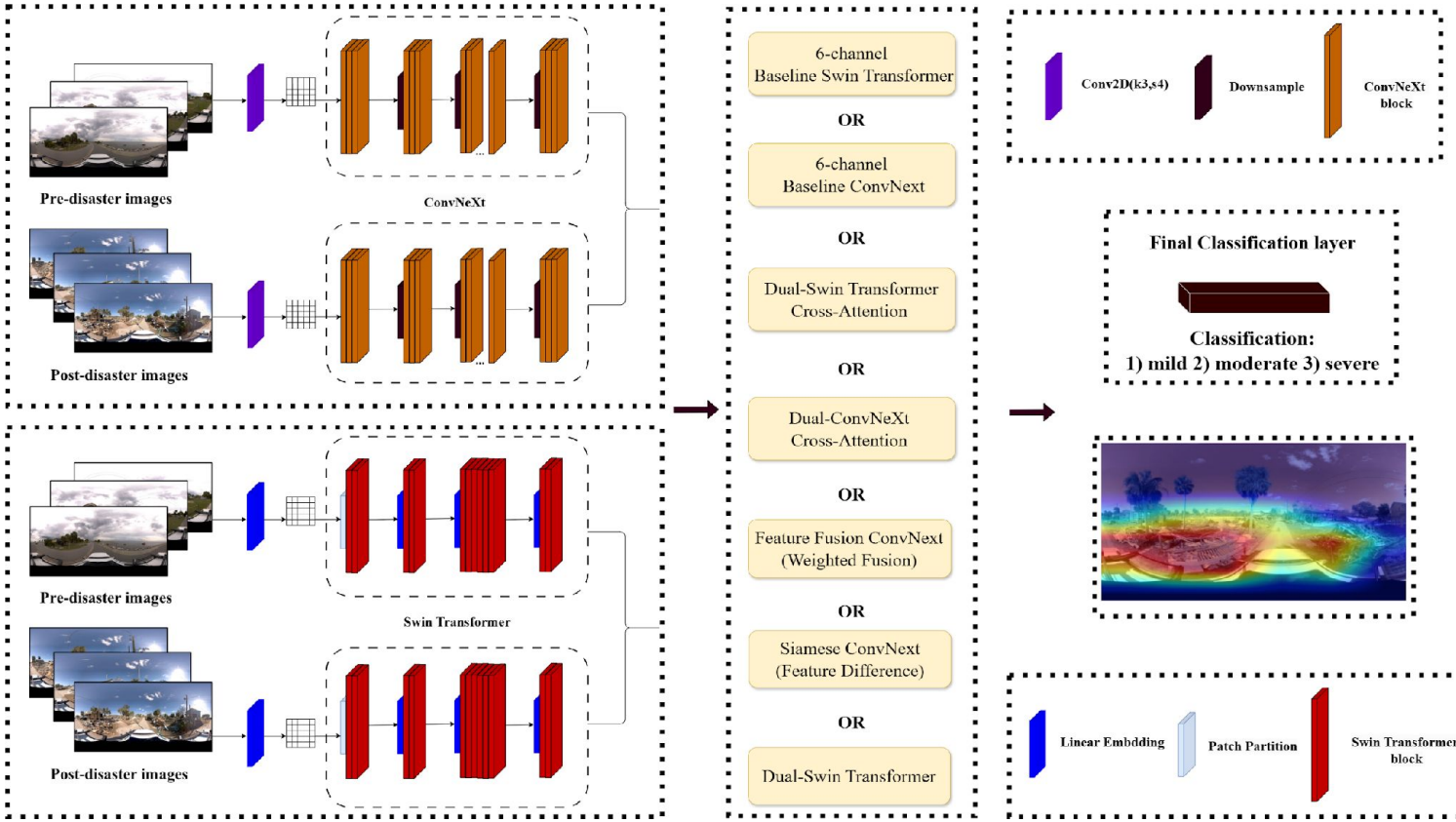
Three settings:

(1) **Post-only**: classify post-event images into 3 levels

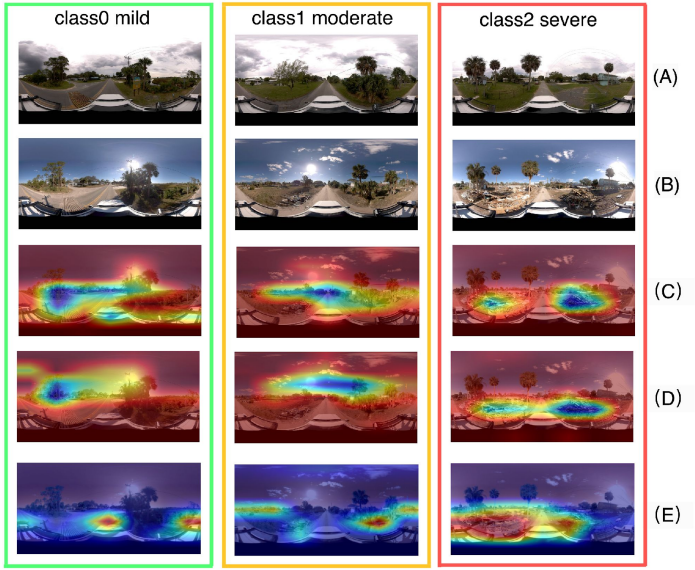
(2) **Pre- and Post-**: pre- and post-event images mixed together, 4 classes (+ no damage)

(3) **Pre- and Post-Paired**: pre- and post-event features fused

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

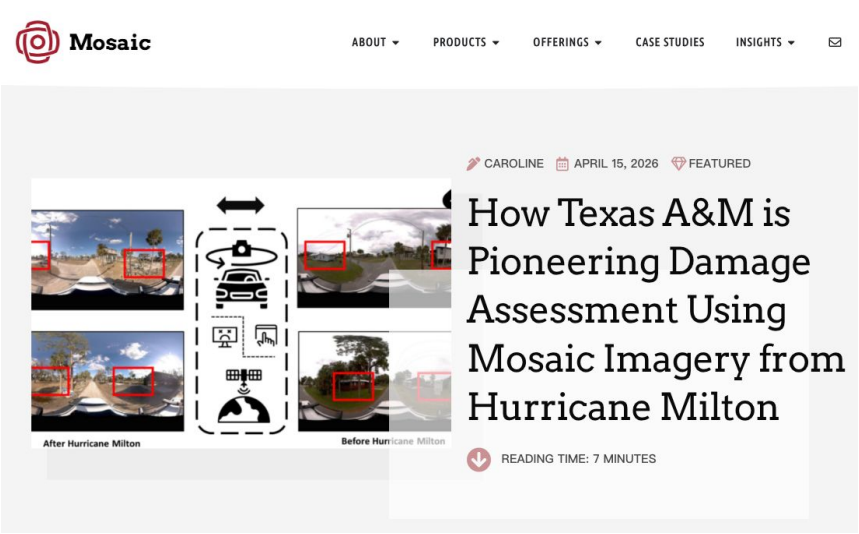


Model name	Accuracy	P	R	F1	Class-Specific Accuracy		
					0-mild	1-moderate	2-severe
Baseline-ConvNext	72.80%	0.7315	0.7280	0.7256	56.91%	79.12%	75.54%
Baseline-Swin Transformer	66.14%	0.6889	0.6614	0.6481	39.02%	84.34%	57.55%
Dual-ConvNeXt + Cross-Attention	76.91%	0.7704	0.7691	0.7672	***71.08%***	72.36%	***91.37%***
Dual-Swin Transformer + Cross-Attention	76.91%	***0.7970***	0.7691	0.7615	51.47%	***91.97%***	74.60%
Dual-Swin Transformer	74.17%	0.7420	0.7417	0.7408	62.69%	75.45%	81.33%
Feature-fusion ConvNext	***77.11%***	0.7751	***0.7710***	***0.7678***	59.39%	82.72%	84.44%
Siamese ConvNeXt	75.54%	0.7878	0.7554	0.7498	53.28%	90.79%	71.11%

Key result: Accuracy improved from 66.14% (Swin baseline) to 77.11% (dual-channel Feature-Fusion ConvNeXt); Grad-CAM confirms pre-disaster images help focus on true pre/post changes.

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)

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https://www.mosaic51.com/featured/how-texas-am-is-pioneering-damage-assessment-using-mosaic-imagery-from-hurricane-milton/

Yang, Y., Zou, L., Zhou, B., Li, D., Lin, B., Abedin, J., & Yang, M. (2025). Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models. Computers, Environment and Urban Systems, 121, 102335.

Computers, Environment and Urban Systems 121 (2025) 102335

Contents lists available at ScienceDirect

Computers, Environment and Urban Systems

journal homepage: www.elsevier.com/locate/ceus

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

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[d] Spatial Sciences Institute, University of Southern California, Los Angeles, USA

ARTICLE INFO

Keywords:
Disaster resilience
Street-view imagery
Dual-channel neural network
Pre-trained vision model
Damage estimation

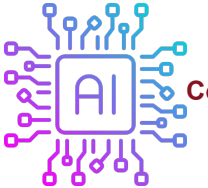
ABSTRACT

Street-view images offer unique advantages for disaster damage estimation as they capture impacts from a visual perspective and provide detailed, on-the-ground insights. Despite several investigations attempting to analyze street-view images for damage estimation, they mainly focus on using post-disaster images. The potential of time-series street-view images remains underexplored. Pre-disaster images provide valuable benchmarks for accurate damage estimations at building and street levels. These images could also aid annotators in objectively labeling post-disaster impacts, improving the reliability of labeled data sets for model training, and potentially enhancing the model performance in damage evaluation. The goal of this study is to estimate hyperlocal, on-the-ground disaster damages using bi-temporal street-view images and advanced pre-trained vision models. Street-view images before and after 2024 Hurricane Milton in Horseshoe Beach, Florida, were collected for experiments. The objectives are: (1) to assess the performance gains of incorporating pre-disaster street-view images as a no-damage category in fine-tuning pre-trained models, including Swin Transformer and ConvNeXt, for damage level classification; (2) to design and evaluate a dual-channel algorithm that reads pair-wise pre- and post-disaster street-view images for hyperlocal damage assessment. The results indicate that incorporating pre-disaster street-view images and employing a dual-channel processing framework can significantly enhance damage assessment accuracy. The accuracy improves from 66.14 % with the Swin Transformer baseline to 77.11 % with the dual-channel Feature-Fusion ConvNeXt model. Gradient-weighted Class Activation Mapping (Grad-CAM) shows that incorporating pre-disaster images improves the pre-trained vision model's capacity to focus on major changes between pre- and post-disaster images, thus enhancing model performance. This research evaluates the technical solutions and challenges of assessing disaster damages at hyperlocal spatial resolutions with bi-temporal street-view images, providing valuable insights to support effective decision-making in disaster management and resilience planning.

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (Under Review, Natural Hazards)

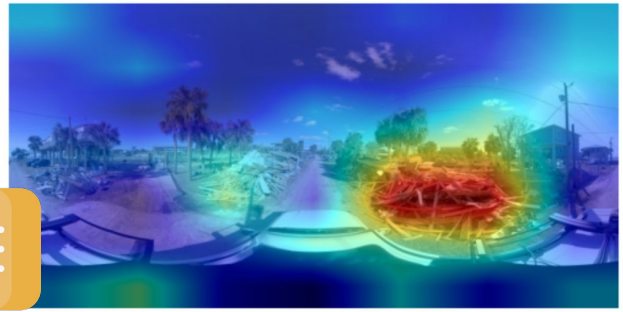


Post-Hurricane
Street-View Imagery



Computer Vision Models

Grad-CAM for Fine-Tuned Swin-Small





Motivation

- Accuracy alone is not enough
- Reliability matters for disaster response
- Overconfident wrong predictions
- Risk of misleading prioritization

Overconfidence=models may give very confident predictions even when they are **wrong**

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

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class0 mild



No buildings in the area were damaged by the hurricane, and the trees appear to have sustained only mild damage. This is truly a relief. The vegetation is lush, the trees are tall, and the damage from the hurricane was minimal. I'd love to live here now, as it's only slightly damaged and looks very livable.

Sagging power lines are visible alongside a clear, dry road. Many trees show defoliation and some branch damage, but no large fallen trees or building debris are present. The overall scene suggests mild environmental impact, with infrastructure largely intact.

class1 moderate



The damage level is moderate, with loose building materials and a small amount of debris suggesting recent construction or minor disrepair. Despite the somewhat cluttered appearance of scattered items, the water features and elevated residences make this area appear to be an attractive residential area.

Scattered wooden debris and loose utility lines on a power pole indicate moderate damage along this coastal dirt road. While buildings appear structurally sound, the presence of scattered planks and beams, alongside the compromised power infrastructure, suggests the area experienced some impact, but not widespread destruction or flooding.

class2 severe



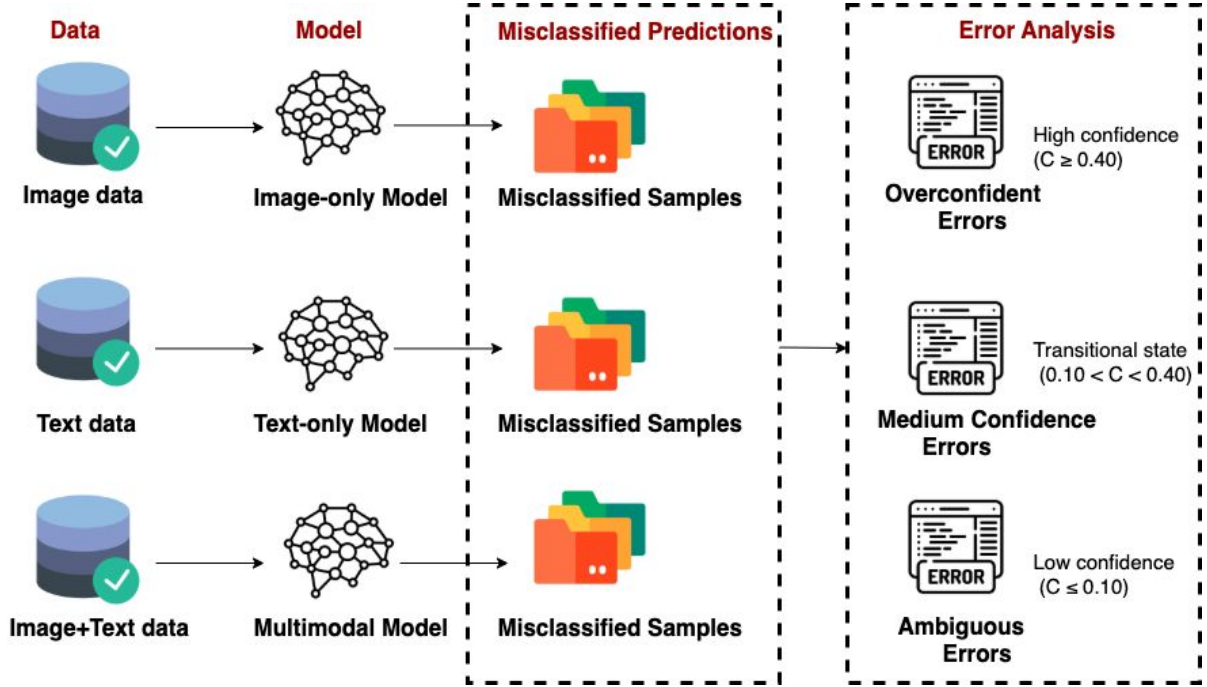
The atmosphere of the image is chaotic, with massive piles of debris covering large parts of the area, giving a strong sense of destruction. The damage appears extensive, with shattered building materials strewn all over, although some elevated houses remain intact in the background.

The street is heavily impacted by extensive building debris, forming large piles along both sides, almost completely blocking passage. This significant accumulation of scattered materials and collapsed fragments suggests severe damage to structures in the vicinity, though visible infrastructure like power lines appears mostly intact.

RQ1: How reliable are image-only, text-only, and image-text models for street-view disaster damage assessment?

RQ2: Can LLM-generated image descriptions support reliable damage assessment when human annotations are limited?

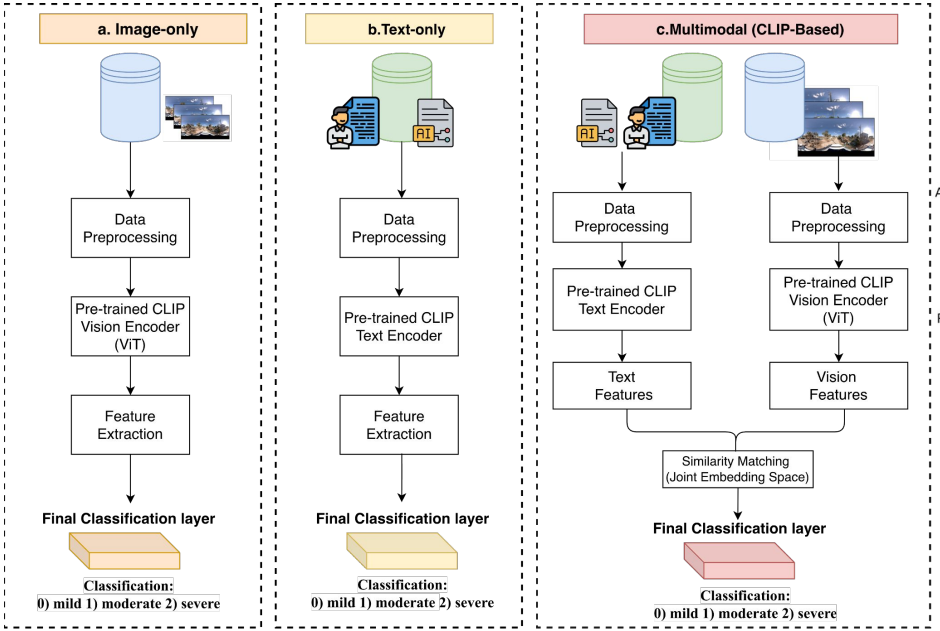
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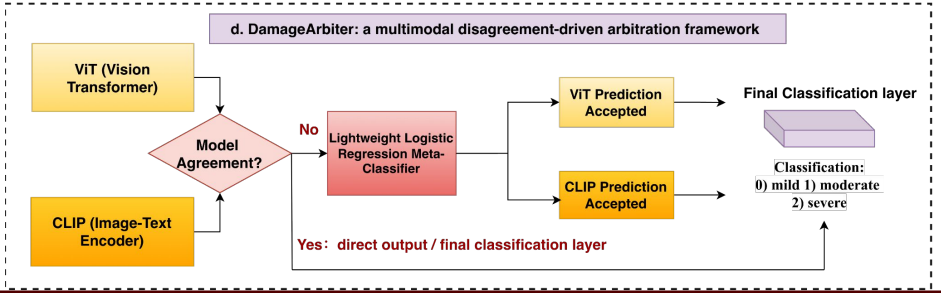


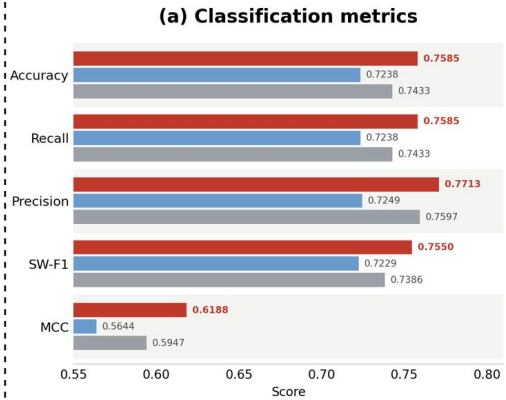
Overconfidence=models may give very confident predictions even when they are wrong

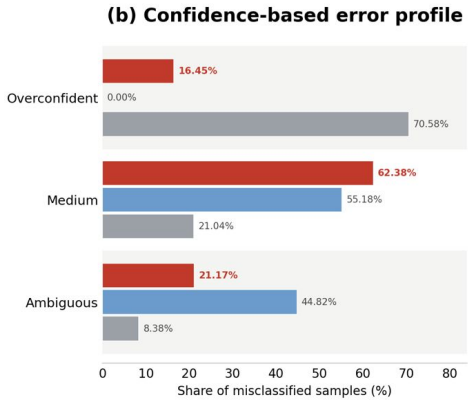
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Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery





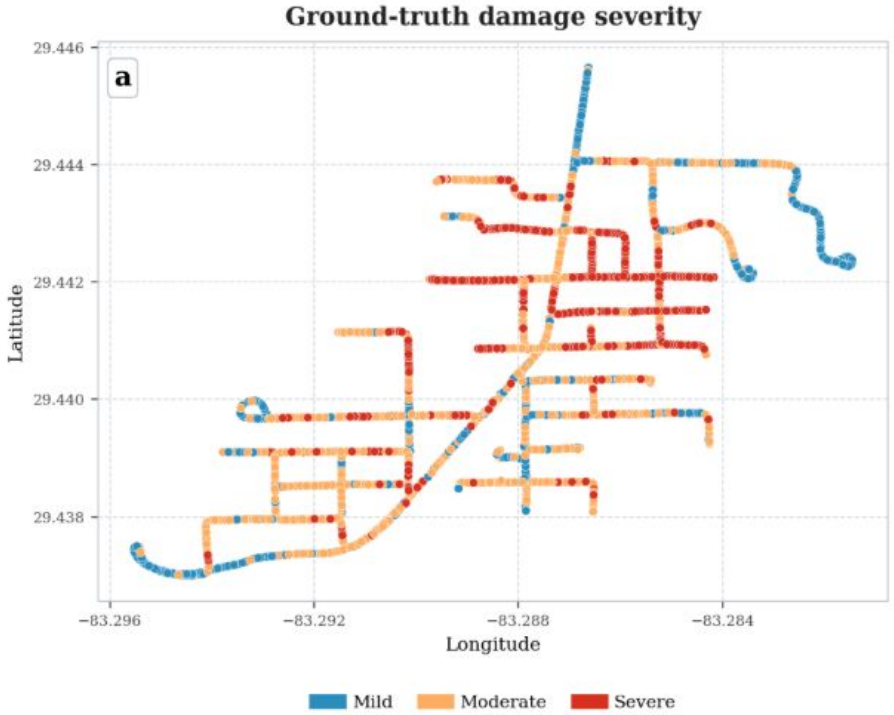


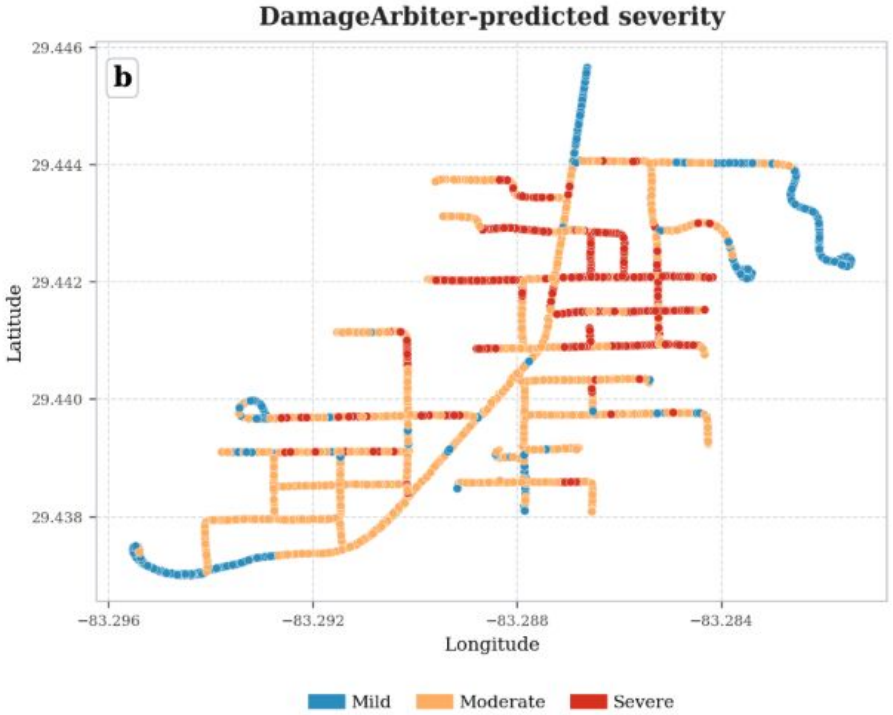


ViT-B/32 CLIP (LLM) DamageArbiter

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Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery





This work won 2nd Place in the 2026 AAG GIS Specialty Group (GISSG) Student Honors Paper Competition.

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STAGE 2 · OBJECTIVE 2

Respond faster

Challenge 2 · Specialized models do not generalize across events

Enable zero-shot assessment that transfers across hazards and regions without new labels or retraining.

Case 3	RAPID: multi-agent pipeline for interpretable cross-view assessment	Accepted · ACM SIGSPATIAL 2026
Case 4	Satellite-to-Street: post-disaster street views from satellite imagery	Accepted · IEEE IGARSS 2026
Case 5	Generative AI for disaster resilience	Under review · CRC Press

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Case 3:

RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (Accepted by ACM SIGSPATIAL)



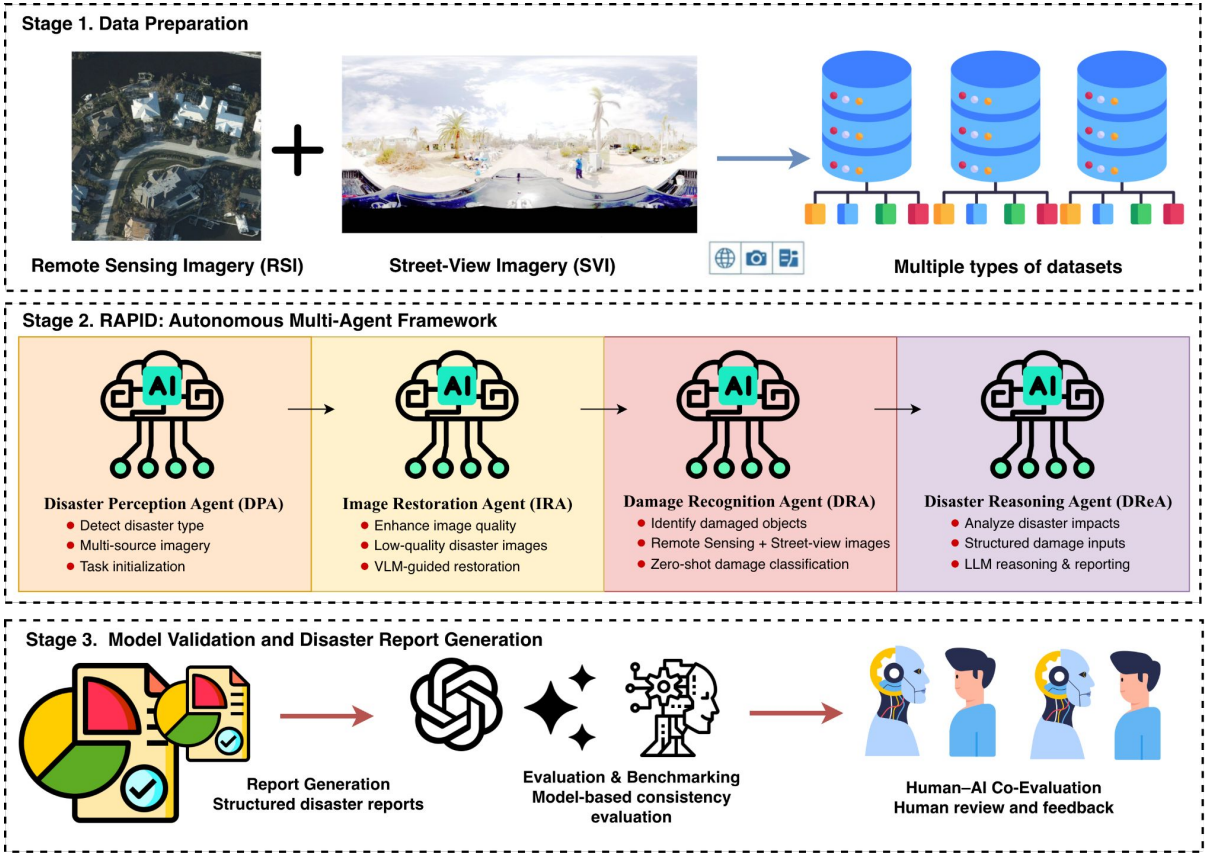
Research Question

Can RAPID support:

- Cross-view assessment?
- Different regions and hazards?
- Rapid zero-shot reasoning?
- Interpretable damage reports?



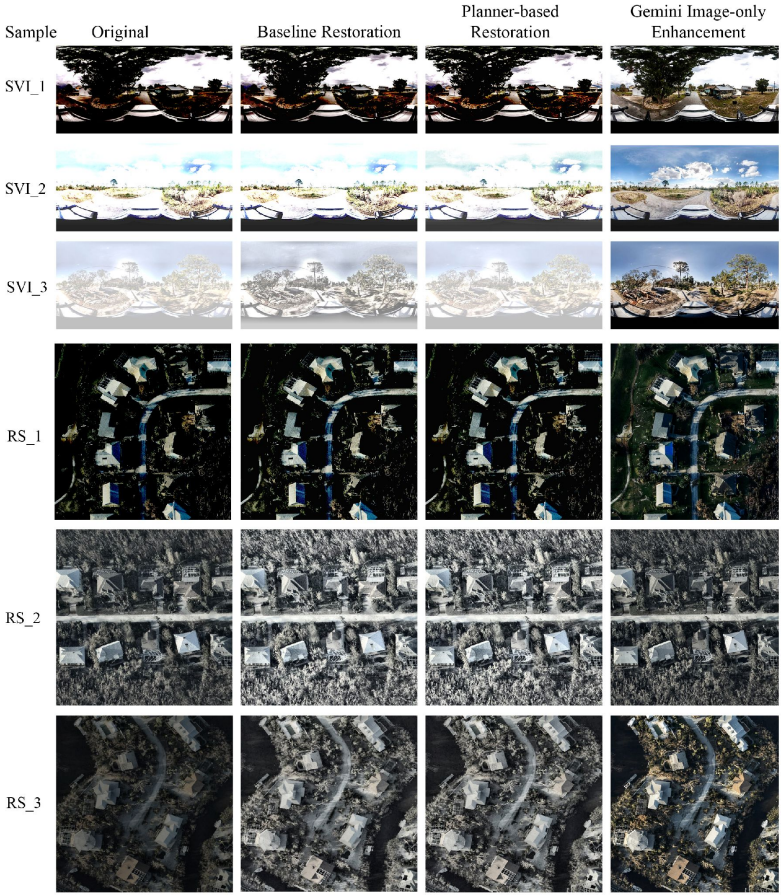
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery



Method:

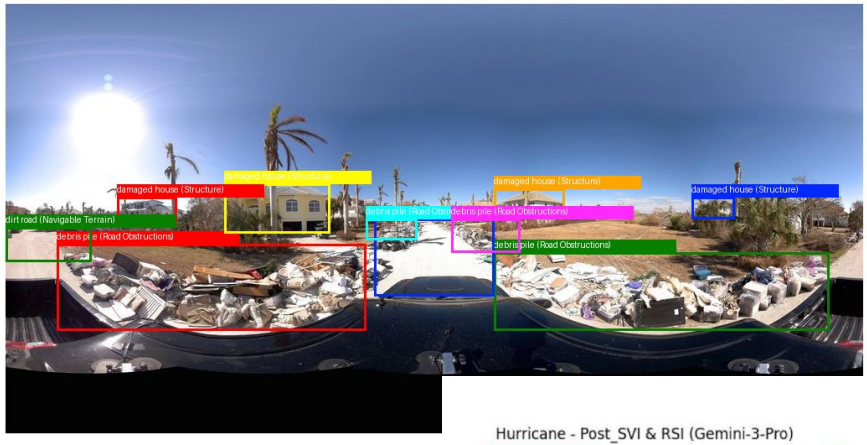
- **Cross-view imagery input**
- **Zero-shot multi-agent assessment**
- **Structured report generation**

Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery



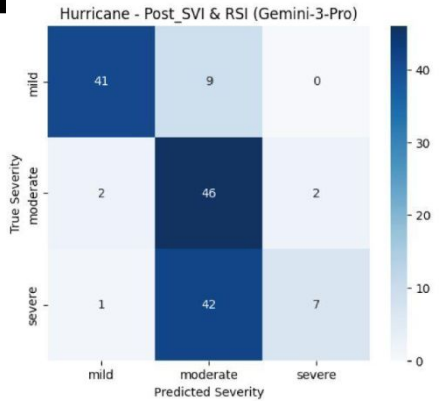
Disaster type	Image type	Restored images	$Q_{original}$	$Q_{baseline}$	$Q_{planner}$	Q_{gemini}
Hurricane	Satellite	141	0.62	**0.73**	0.71	0.69
Hurricane	SVI	22	0.75	0.78	0.76	**0.79**
Hurricane	SVI	4	0.76	0.78	0.78	**0.79**
Wildfire	SVI	14	0.61	**0.67**	**0.67**	0.57
Hurricane	SVI	2	0.51	0.55	**0.62**	0.60
Ice storm	SVI	4	0.73	0.74	**0.75**	0.67
Flood	SVI	2	0.68	0.73	**0.77**	0.71
Earthquake	SVI	2	0.71	0.79	**0.80**	0.77
Drought	SVI	2	0.66	0.75	**0.78**	0.61
Wildfire	SVI	3	0.70	0.77	**0.78**	**0.78**

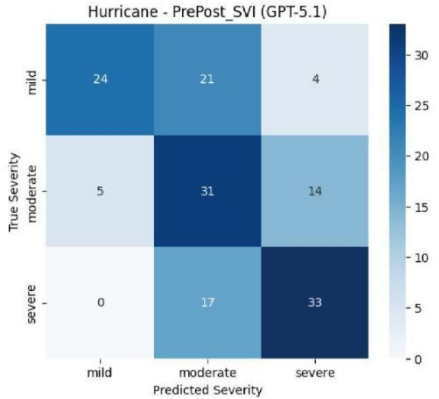
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

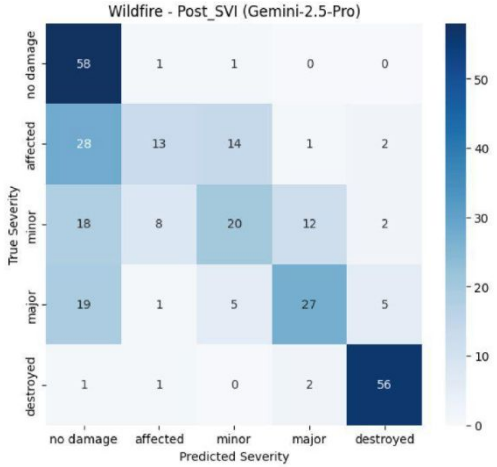


Metric	Model	Dataset A	Dataset B	Dataset C2
Accuracy ↑	GPT-5-mini	0.387	0.503	0.573
	ChatGPT-5.1	0.573	**0.591**	0.570
	Gemini-2.5-flash	0.360	0.470	0.559
	Gemini-2.5-Pro	0.380	0.447	**0.590**
	Gemini-3-Pro	**0.627**	0.493	0.442
NCSE ↓	GPT-5-mini	0.307	0.248	0.165
	ChatGPT-5.1	0.213	**0.218**	0.173
	Gemini-2.5-flash	0.363	0.299	0.179
	Gemini-2.5-Pro	0.373	0.327	**0.162**
	Gemini-3-Pro	**0.190**	0.291	0.210



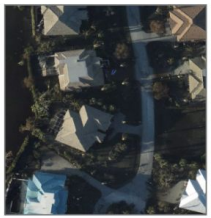
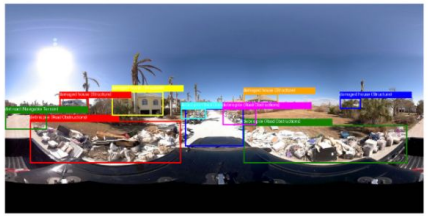






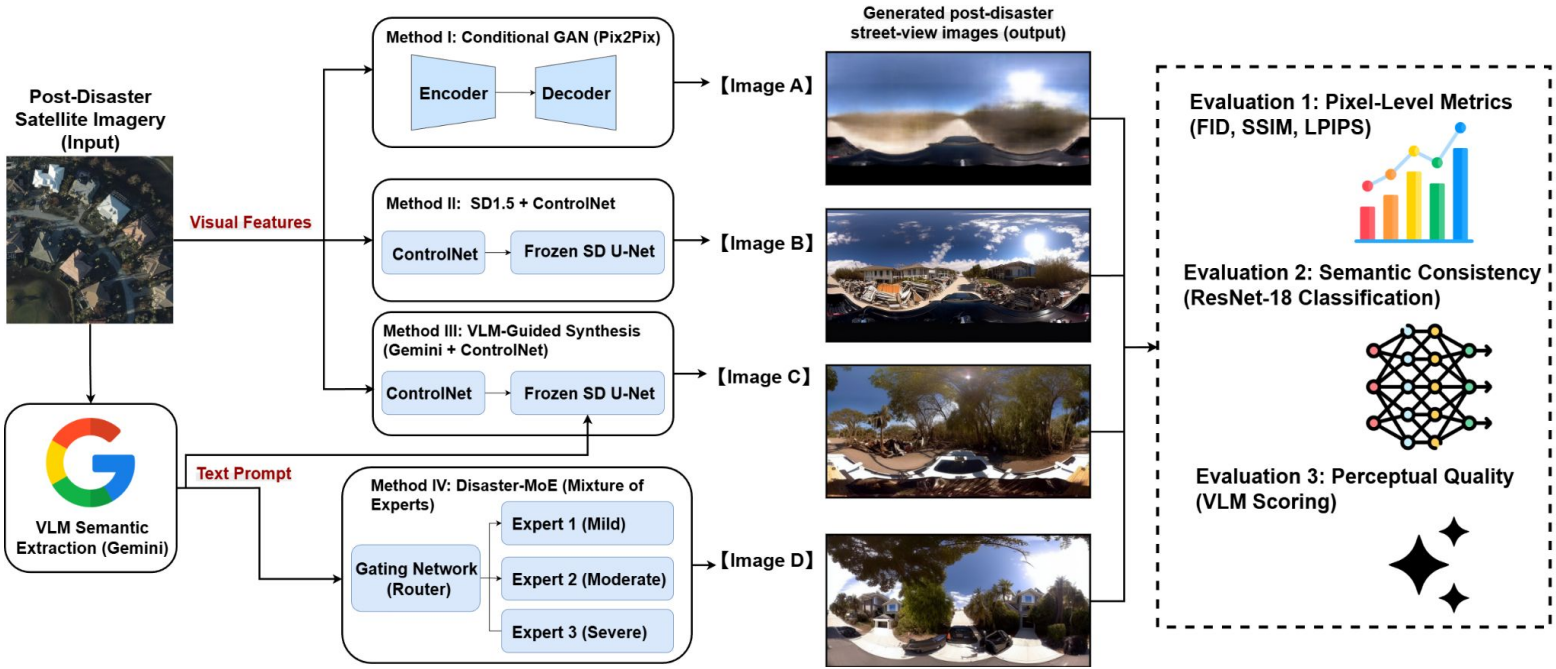
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
 	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building
 Using Gemini-3-Pro for object recognition	The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. **Reasoning for Gemini-3-Pro-Disaster Assessment**			
	Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. **Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion**			

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Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)



RQ1: Can generative vision models synthesize realistic post-disaster street views from satellite imagery?
RQ2: Do explicit disaster semantics improve synthesis fidelity?
RQ3: How can we reliably evaluate cross-view disaster synthesis beyond pixel metrics?

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Case 5: Generative AI for Disaster Resilience

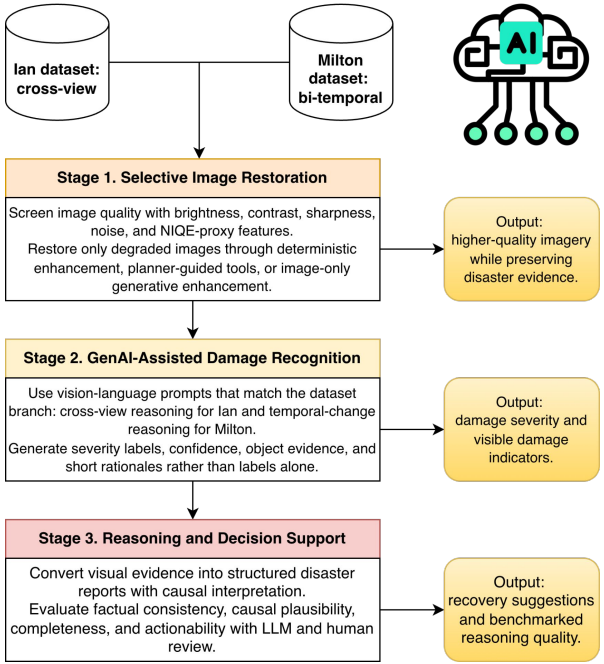
(Under Review, Geography in the Age of Generative AI (CRC Press))

Generative AI Across the Disaster Resilience Lifecycle

Geospatial data inputs: Remote sensing, Street-view imagery, Sensors, Social media, Infra/DEM, Field surveys

	Mitigation	Preparedness	Response	Recovery
Challenge	Risk identification requires static baseline data, nonlinear hazard-vulnerability interactions, and future scenario assumptions.	Monitoring must handle fast-changing data streams, missing observations, and uncertain event evolution before impacts occur.	Damage assessment and emergency coordination occur under time pressure, incomplete data, and rapidly changing priorities.	Recovery planning must compare uncertain reconstruction paths, long-term trade-offs, and uneven community needs.
GenAI support	Generate plausible hazard scenarios; align terrain, land-use, exposure, and historical records; reason over cross-scale risk drivers.	Complete noisy streams, forecast evolving hazards, and simulate warning, evacuation, and resource-allocation plans.	Restore degraded imagery, classify severity with generated evidence, and coordinate agency-specific decisions through agent roles.	Generate recovery scenarios, reason about trade-offs, and support participatory planning with human oversight.
Output	Scenario-based risk maps and mitigation options	Early warning and preparedness plans	Damage maps and emergency actions	Adaptive rebuilding and resilience strategies

Core GenAI capabilities: generation | multimodal understanding | reasoning | multi-agent collaboration



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STAGE 3 · OBJECTIVE 3

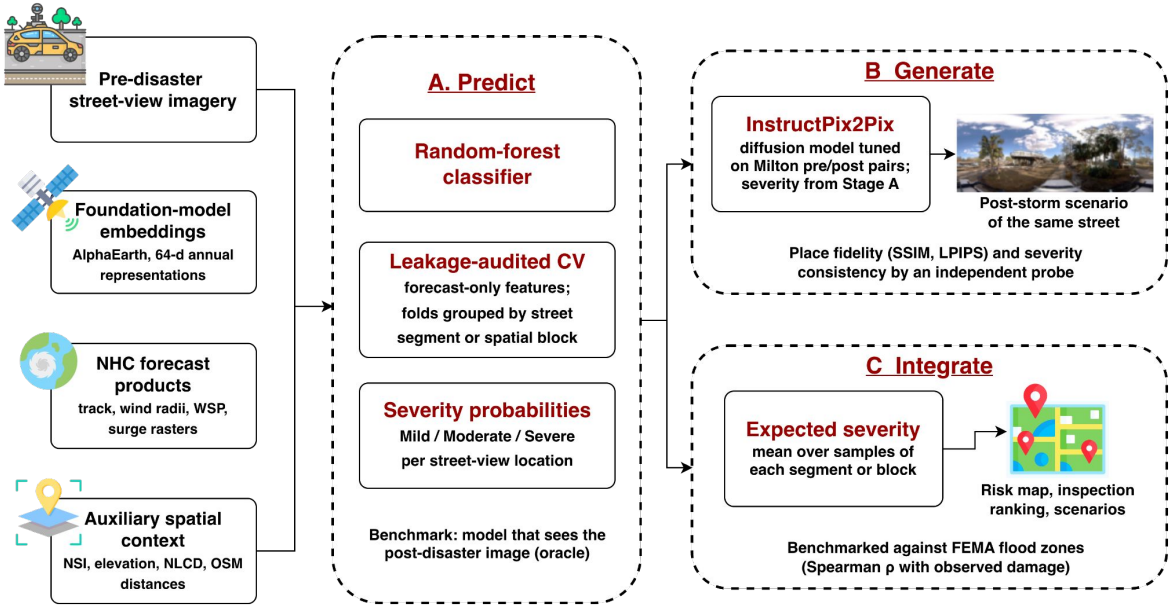
Prepare and deliver better

Challenge 3 · Assessment alone does not close the resilience loop

Build accountable agentic GeoAI that turns assessment into evidence-backed action before, during, and after disasters.

Case 6	Seeing damage before it happens: anticipatory street-level prediction	Ongoing
Case 7	RAY: an accountable GeoAI system for place-based disaster intelligence	Ongoing

Case 6: Seeing Disaster Damage Before It Happens: Predicting and Visualizing Street-Level Impacts from Multimodal Pre-Event Information (Ongoing)



Case studies: Hurricane Milton 2024, Horseshoe Beach: Stages A, B, C Hurricane Ian 2022, Lee County: Stages A, C

Research Question:

- **Can we anticipate post-disaster street-level damage using pre-disaster multi-modal data, Earth observation embeddings, and hazard forecasts?**

Objective:

- **To develop a preparedness-oriented GeoAI framework that transforms pre-disaster data and forecasts into anticipatory damage predictions, spatial risk prioritization, and scenario visualizations.**

Case 7: RAY: Resilience Assistant for You
An Accountable GeoAI System for Place-Based Disaster Intelligence (Ongoing)

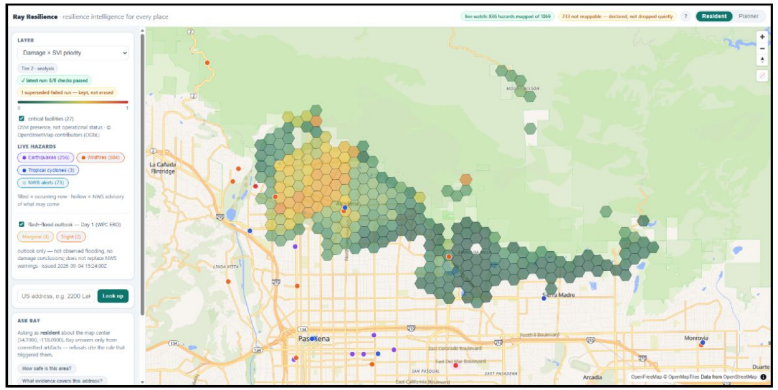
Data plane

GitHub Actions, hourly + on push
- Tier-1 watch: USGS · NWS · NHC · NIFC
- Deep-case builders: Eaton, Milton, Ian
- Every artifact: SHA-256 + audit row
- Append-only snapshots

Agent plane (Ray)

Any OpenAI-compatible endpoint
pre-check → grounded generation
→ citation post-check → audit
- Cites |artifact:...| or refuses
- Every refusal names its rule

Presentation plane



Steward Harness

CI gate: the assembled public site is verified against the distribution plane; a violating deploy fails

Outcome validity
Executable spatial checks: CRS, join integrity, bounds. Per-tile uncertainty is mandatory, or the build fails.

Process validity
SHA-256 per artifact; append-only audit rows (agent, inputs, UTC). Failures recorded, never erased.

Institutional validity
One YAML policy, two planes: claim = what may be asserted; distribution = what may ship. Ordered rules, default deny.

Verifiability
retained > re-derivable > cited-only; weakest link. A license attribute gates redistribution by rule.

Research Question:

How can an **automated, multi-agent GeoAI system** deploy large models within WebGIS and mobile apps to deliver accountable, evidence-backed, and real-time resilience awareness for both the public and decision-makers?

Objective:

To implement an **accountable multi-agent GeoAI architecture**—such as the RAY system—that transforms complex spatial data into real-time, interactive insights featuring explicit evidence lineage, planning trade-off evaluation, and differentiated multi-tier dissemination.

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Publications and Awards

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As of October 2026 · Google Scholar: 434 citations, h-index 8, i10-index 7.

PUBLICATIONS

8 first-authored publications during Ph.D.

- 2 journal articles (CEUS 2025; Natural Hazards under review)
- 5 conference proceedings (ACM SIGSPATIAL 2026; ACM SIGSPATIAL Workshop GeoSearch 26; IEEE IGARSS 2026; CaGIS 2026; ICC 2025)
- 1 chapter under review (CRC Press)

12 co-authored publications

- 6 journal articles (e.g., Journal of Hazardous Materials, Int. J. Applied Earth Observation and Geoinformation, Urban Informatics, Journal of Transport Geography)
- 3 conference papers (ACM SIGSPATIAL 2026 short paper; AGU 2025) and 3 preprints



HONORS AND AWARDS

A total of 15 honors received during Ph.D.

2026

- 2nd Place, AAG GISSG Student Honors Paper Competition
- AAG International Geographic Information Fund (IGIF) Scholarship
- CaGIS Doctoral Scholarship Award
- IGARSS 2026 Travel Grant
- NSF I-GUIDE Summer School Travel Award
- ACM SIGSPATIAL 2026 Travel Award
- Geography Graduate Student Enhancement Scholarship, Department of Geography, etc.

2025

- Best Student Paper Award, International Cartographic Conference (ICC 2025)
- Environment and Sustainability Graduate Fellow Award, TAMU
- TAMIDS Student Ambassador Scholarship
- CaGIS International Travel Grant, etc.

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Research Schedule

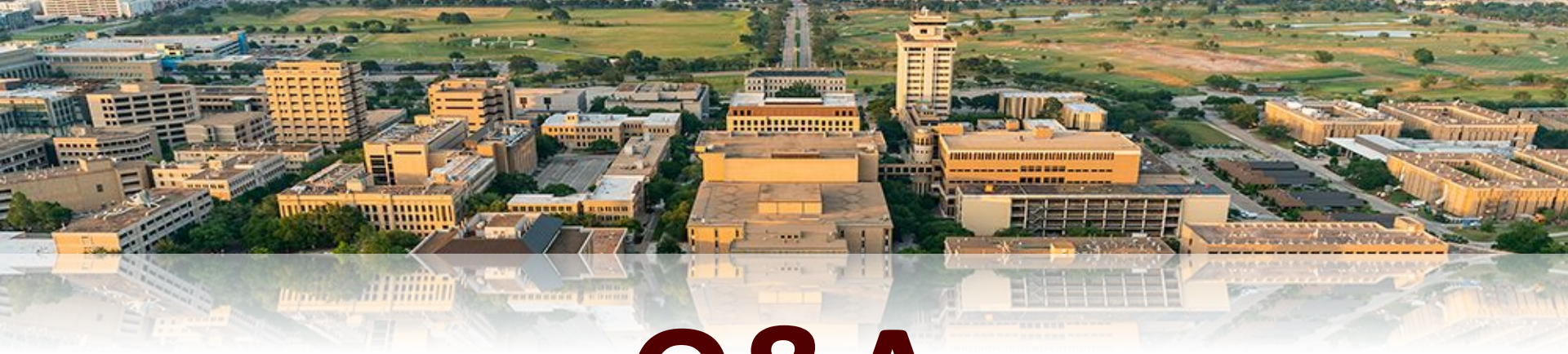
Now

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Tasks	Fall 2024	Spring 2025	Summer 2025	Fall 2025	Spring 2026	Summer 2026	Fall 2026	Spring 2027	Summer 2027
Case Study 1	Done								
Case Study 2		Done							
Case Study 3				Done					
Case Study 4					Done				
Case Study 5					Done				
Case Study 6							In Progress (20%)		
Case Study 7							In Progress (40%)		
Prelim Exam					Done				
Proposal Defense							In Progress		
Dissertation Writing							In Progress		
Final Defense & Graduation									Planned

Done In Progress Planned

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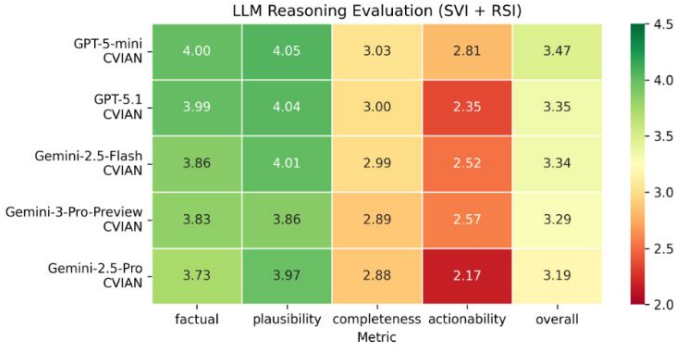
Q&A

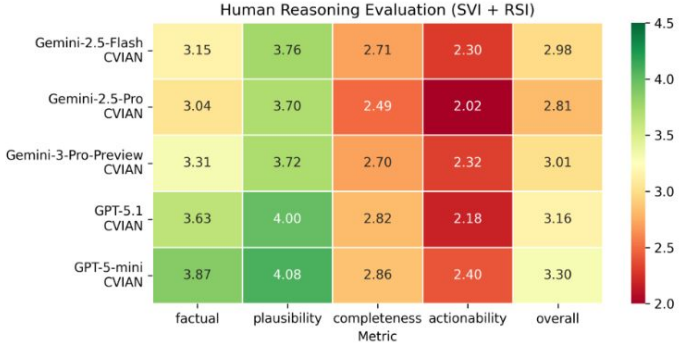
Welcome any comments/questions

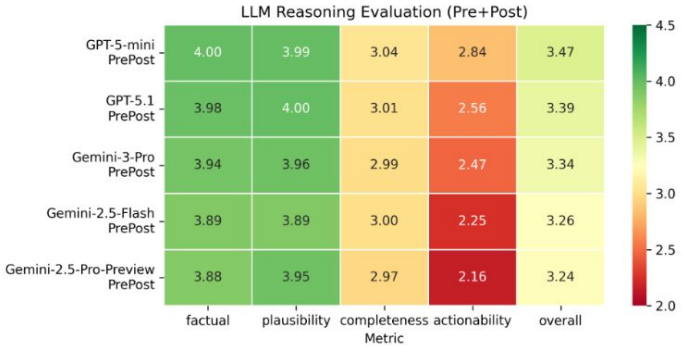
yyang295@tamu.edu

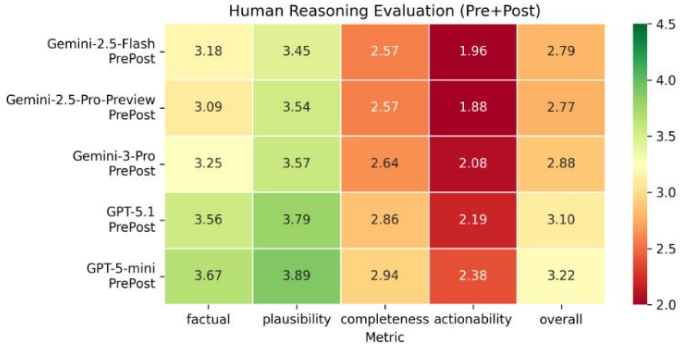
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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

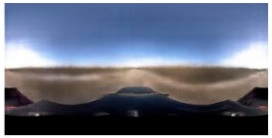
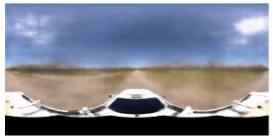
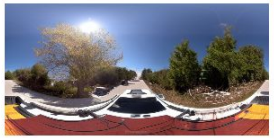








Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)

	Satellite	Pix2Pix	SD1.5+ControlNet	SD1.5+ControlNet+Gemini	MoE	Ground Truth
Mild						
Moderate						
Severe						

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Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)

Table 1 Quantitative Comparison of Perceptual Image Quality Metrics

Method	SSIM(↑)	PSNR(↑)	LPIPS(↓)	FID(↓)
Pix2Pix	**0.58609614**	**15.30729573**	**0.549302574**	150.8341348
SD1.5_ControlNet	0.31436218	9.809302381	0.601944895	**74.32713682**
SD1.5_ControlNet_Gemini	0.290885676	9.726198367	0.603852578	82.19407033
MoE	0.221760725	8.453305609	0.6879829	134.522164
LoRA	0.1312	8.9619	0.6223	171.5578

Table 2 Classification Accuracy Score for Disaster Severity Consistency

Method	Accuracy	Recall	Precision	F1_Score	Mild_Acc	Mod_Acc	Sev_Acc
GroundTruth	*0.73*	*0.73*	*0.76*	*0.74*	*0.77*	*0.76*	*0.66*
Pix2Pix	0.34	0.34	0.28	0.17	**1**	0.01	0
SD1.5_ControlNet	**0.72**	**0.72**	**0.74**	**0.71**	**0.91**	0.4	**0.86**
SD1.5_ControlNet_Gemini	0.43	0.43	0.45	0.43	0.4	0.39	0.5
MoE	0.43	0.43	0.45	0.44	0.41	**0.47**	0.42
LoRA	0.37	0.37	0.39	0.34	0.27	0.12	0.73

- Single-model approaches can't yet balance visual plausibility with strict structural alignment
- High perceptual realism ≠ high semantic accuracy — evaluation must be structure-aware
- Trustworthiness, not fidelity alone, should be the target for disaster generation